

Vehicle Setup Guide (Manual, from scratch)

This guide describes how to add a new car (example: Mercedes) so it appears in Main Menu and works in game exactly like existing cars.

1) Create folder structure

Create:

- Assets/CarData/Vehicles/<YourCar>/Prefabs/Body
- Assets/CarData/Vehicles/<YourCar>/Prefabs/Wheels
- Assets/CarData/Vehicles/<YourCar>/ScriptableObjects/Config
- Assets/CarData/Vehicles/<YourCar>/ScriptableObjects/Tuning
- Assets/CarData/Vehicles/<YourCar>/ScriptableObjects/Paints
- Assets/CarData/Vehicles/<YourCar>/ScriptableObjects/BodySets

Fastest template path: duplicate Assets/CarData/Vehicles/Buick_GNX and rename files/assets.

2) Prepare body prefabs

In Prefabs/Body prepare:

- Main body prefab (Car_<YourCar>_SetA.prefab recommended)
- Optional menu body prefab (Car_<YourCar>_Menu.prefab) if your pipeline uses separate menu visuals
- Body set prefabs (BodySetA.prefab, BodySetB.prefab, ...)

Expected hierarchy in main body prefab:

- Body
- Root (common meshes that are always present)
- Body sets are instantiated as siblings near Root at local zero transform.

For each body set prefab:

- localPosition = (0,0,0)
- localRotation = (0,0,0)
- localScale = (1,1,1)

3) Prepare wheel prefab

In Prefabs/Wheels create wheel visual prefab used by PlayerCarConfig.visual.wheelPrefab.

4) Create ScriptableObjects

Create these assets (or duplicate Buick ones and retarget):

4.1 PlayerCarConfig

Path example:

- ScriptableObjects/Config/<YourCar>.asset

Fill at minimum:

- `visual.bodyPrefab` -> your main body prefab
- `visual.wheelPrefab` -> your wheel prefab
- `visual.generateConvexBodyColliders` as needed
- wheelbase/axle/wheel height values for your model
- damage section (can start by copying Buick values)

4.2 BodySetConfig assets

Path example:

- ScriptableObjects/BodySets/<YourCar>_BodySet_SetA.asset
- ScriptableObjects/BodySets/<YourCar>_BodySet_SetB.asset

Set:

- `BodySetPrefab` -> corresponding BodySetX.prefab

4.3 Tuning assets

Create/duplicate:

- VehicleHandlingConfig
- several EngineConfig

- several SuspensionConfig

4.4 Paint assets

Create multiple paint option assets for garage selection.

4.5 CarLoadoutConfig

Path example:

- ScriptableObjects/Tuning/<YourCar>_Loadout.asset

Set fields:

- displayName
- playerCarConfig
- handlingConfig
- bodySets[]
- engineConfigs[]
- suspensionConfigs[]
- paintOptions[]
- defaults indexes

5) Connect to Main Menu

Open Assets/Scenes/MainMenu.unity.

Find GarageMenuController component and append your loadout into:

- carLoadouts

After this, car appears in menu carousel/list.

6) Validation checklist

1. Car appears in Main Menu selection.
2. Selecting car spawns correct body + wheels in preview.
3. Body set switch in garage replaces set prefab correctly.
4. Start game spawns selected car and selected body set.
5. Damage/deformation applies visually on chosen body.

7) What to retarget first after cloning Buick

If you duplicated Buick as a template, replace these references first:

- PlayerCarConfig.visual.bodyPrefab
- PlayerCarConfig.visual.wheelPrefab
- Every BodySetConfig.bodySetPrefab
- CarLoadoutConfig.displayName

Then tune handling/suspension values.